

Estéban Gabory

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Scientific employment

- 2025 – Now • **Postdoc** Wrocław, Poland,
Supervisor: Paweł Gawrychowski
- 2024 – 2025 • **Assegno di Ricerca (Post-doctoral position)** Palermo, Italy,
Supervisor: Gabriele Fici,
Title: *Combinatorics on words for machine learning*.

Higher education

- 2021 – 2024 • **Ph.D. in Computer Science**, CWI Amsterdam, The Netherlands,
Supervisors: Solon P. Pissis, Leen Stougie,
Thesis title: *Variable strings for pangenomes: Matching, Comparison, Alignment*.
- 2020 – 2021 • **M.Sc. Fundamental Mathematics, Université de Paris** (Second year) Paris, France,
Supervisor: Laurent Bulteau,
Thesis title: *Parametrized algorithms for consensus problems with swaps*.
- 2020 • **Summer research internship, Université de Brest** Brest, France,
Supervisor: Johannes Huisman,
Topic: *Class Field Theory*.
- 2019 – 2020 • **M.Sc. Fundamental Mathematics, Université Rennes 1** (First year) Rennes, France,
Supervisor: Ludovic Marquis,
Thesis title: *Tessellations of the Hyperbolic Plane*.
- 2018–2019 • **Exchange year in the department of Mathematics, Sapienza Università di Roma**
Rome, Italy.
- 2017 – 2019 • **Bachelor in Mathematics, Sorbonne Université** Paris, France.

Professional service

Organising Committee Memberships

- Symposium on String Processing and Information Retrieval (SPIRE 2023).
- European Symposium on Algorithms (ALGO 2023).

Program Committee Memberships

- International Symposium on String Processing and Information Retrieval (SPIRE 2026)
- IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2023).

Reviews for conferences

- European Symposium on Algorithms (ESA 2026) - **2 papers**.
- Workshop on Algorithms in Bioinformatics (WABI 2026).
- International Colloquium on Automata, Languages, and Programming (ICALP 2026).
- Symposium on Theory of Computing (STOC 2026).
- International Conference on Developments in Language Theory (DLT 2026).
- Prague Stringology Conference (PSC 2025).

- Annual Symposium on Combinatorial Pattern Matching (CPM 2025).
- Symposium on Experimental Algorithms (SEA 2025).
- Workshop on Algorithms in Bioinformatics (WABI 2024).
- Annual Symposium on Combinatorial Pattern Matching (CPM 2024).
- Symposium on String Processing and Information Retrieval (SPIRE 2023).
- Annual Symposium on Combinatorial Pattern Matching (CPM 2023).
- Workshop on Algorithms in Bioinformatics (WABI 2022).

Reviews for journals

- Theory of computing systems.
- Algorithms for Molecular Biology - **2 papers**.
- Information and Computation - **3 papers**.

Teaching

Università di Palermo

- Let's create an app (Editions in 2024 and 2025) - Introduction to JavaScript app creation for high school students

Vrije Universiteit

- Algorithms in Sequence Analysis, Master Course (2023) - **Teaching Assistant**.

Université de Paris

- Undergraduate Algebra, third year (2021) - **Teaching Assistant**.
- Undergraduate Algebra, second year (2021) - **Teaching Assistant**.
- Undergraduate Analysis, third year (2021) - **Teaching Assistant**.
- Undergraduate Analysis, second year (2021) - **Teaching Assistant**.

Association Science Technologie et Santé

- Introduction to programming with "Scratch" - Elementary school supervision for scientific activities.

Talks at international conferences

- CPM 2026: "Totally Unclustered BWT Images of Any Length over Non-Binary Alphabets", Copenhagen, Denmark, June 15th, 2026
- SPIRE 2025: "String Consensus Problems with Swaps and Substitutions", London, UK, September 8th, 2025
- ICDE 2024: "Space-Efficient Indexes for Uncertain Strings", Utrecht, The Netherlands, May 17th, 2024
- CPM 2023: "Comparing Elastic-Degenerate Strings: Algorithms, Lower Bounds, and Applications", Marne-la-Vallée, France, June 26th, 2023
- LATIN 2022: "Elastic-Degenerate String Matching with 1 Error", Guanajuato, Mexico, November 10th, 2022
- CPM 2022: "On Strings Having the Same Length-k Substrings", Prague, Czechia, February 22nd, 2022

PhD Courses, schools, and workshops

- Landelijk Netwerk Mathematische Besliskunde - "Randomized Algorithms" (2023).
- Winter School - "Collaborating with wet-lab biologists" (2023).
- Landelijk Netwerk Mathematische Besliskunde - "Networks and Semidefinite Programming" (2022).

- Annual ALPACA Workshop - “Pan-genome sequencing and analysis” (2022).
- Summer School - “Algorithmic techniques in pan-genomics” (2022).
- Landelijk Netwerk Mathematische Besliskunde - “Integer Programming Methods” (2021).
- Annual ALPACA Workshop - “Challenges of computational pan-genomics” (2021).

Research Grants

- Marie Skłodowska-Curie Innovative Training Network (MSCA-ITN-2020) on ALgorithms for PAngenome Computational Analysis (ALPACA) – Role: Early Stage Researcher.

Publications

In international peer-reviewed conference proceedings

- [1] Itai Boneh, Estéban Gabory, Paweł Gawrychowski, and Adam Górkiewicz. “Balancing Two-Dimensional Straight-Line Programs”. *37th Annual Symposium on Combinatorial Pattern Matching (CPM 2026)*  DOI: 10.4230/LIPIcs.CPM.2026.32 (2026).
- [2] Gabriele Fici, Estéban Gabory, Giuseppe Romana, and Marinella Sciortino. “Totally Unclustered BWT Images of Any Length over Non-Binary Alphabets”. *37th Annual Symposium on Combinatorial Pattern Matching (CPM 2026)*  DOI: 10.4230/LIPIcs.CPM.2026.13 (2026).
- [3] Estéban Gabory, Laurent Bulteau, Gabriele Fici, and Hilde Verbeek. “String Consensus Problems with Swaps and Substitutions”. *String Processing and Information Retrieval* (2026).
- [4] Gabriele Fici and Estéban Gabory. “Generalized De Bruijn Words, Invertible Necklaces, and the Burrows-Wheeler Transform”. *50th International Symposium on Mathematical Foundations of Computer Science (MFCS 2025)*  DOI: 10.4230/LIPIcs.MFCS.2025.48 (2025).
- [5] Rocco Ascone, Giulia Bernardini, Alessio Conte, Massimo Equi, Estéban Gabory, Roberto Grossi, and Nadia Pisanti. “A Unifying Taxonomy of Pattern Matching in Degenerate Strings and Founder Graphs”. *24th International Workshop on Algorithms in Bioinformatics (WABI 2024)*  DOI: 10.4230/LIPIcs.WABI.2024.14 (2024).
- [6] Esteban Gabory, Chang Liu, Grigorios Loukides, Solon P. Pissis, and Wiktor Zuba. “Space-Efficient Indexes for Uncertain Strings”. *2024 IEEE 40th International Conference on Data Engineering (ICDE)*  DOI: 10.1109/ICDE60146.2024.00367 (2024).
- [7] Estéban Gabory, Eric Rivals, Michelle Sweering, Hilde Verbeek, and Pengfei Wang. “Periodicity of Degenerate Strings”. *Proc. of 26th Prague Stringology Conference*. ISBN: 978-80-01-07206-6. Faculty of Information Technology, Czech Technical University in Prague (2023).
- [8] Esteban Gabory, Moses Njagi Mwaniki, Nadia Pisanti, Solon P. Pissis, Jakub Radoszewski, Michelle Sweering, and Wiktor Zuba. “Comparing Elastic-Degenerate Strings: Algorithms, Lower Bounds, and Applications”. *34th Annual Symposium on Combinatorial Pattern Matching (CPM 2023)*  DOI: 10.4230/LIPIcs.CPM.2023.11 (2023).
- [9] Giulia Bernardini, Esteban Gabory, Solon P. Pissis, Leen Stougie, Michelle Sweering, and Wiktor Zuba. “Elastic-Degenerate String Matching with 1 Error”. *LATIN 2022: Theoretical Informatics*  DOI: 10.1007/978-3-031-20624-5_2 (2022).
- [10] Giulia Bernardini, Alessio Conte, Esteban Gabory, Roberto Grossi, Grigorios Loukides, Solon P. Pissis, Giulia Punzi, and Michelle Sweering. “On Strings Having the Same Length-k Substrings”. *33rd Annual Symposium on Combinatorial Pattern Matching (CPM)*  DOI: 10.4230/LIPIcs.CPM.2022.16 (2022).

In international peer-reviewed journals

- [11] Rocco Ascone, Giulia Bernardini, Alessio Conte, Massimo Equi, Esteban Gabory, Roberto Grossi, and Nadia Pisanti. “Pattern Matching with Elastic-Degenerate Strings and Elastic-Founder Graphs”. *Algorithms for Molecular Biology*  DOI: 10.1186/s13015-025-00289-3 (2026).
- [12] Giulia Bernardini, Alessio Conte, Esteban Gabory, Roberto Grossi, Grigorios Loukides, Solon P. Pissis, Giulia Punzi, and Michelle Sweering. “On Strings Having the Same Length-k Substrings”. *Theory of Computing Systems*  DOI: 10.1007/s00224-025-10240-z (2026).
- [13] Estéban Gabory, Njagi Moses Mwaniki, Nadia Pisanti, Solon P. Pissis, Jakub Radoszewski, Michelle Sweering, and Wiktor Zuba. “Elastic-degenerate string comparison”. *Inf. Comput.*  DOI: 10.1016/J.IC.2025.105296 (2025).
- [14] Esteban Gabory, Moses Njagi Mwaniki, Nadia Pisanti, Solon P. Pissis, Jakub Radoszewski, Michelle Sweering, and Wiktor Zuba. “Pangenome comparison via ED strings”. *Frontiers in Bioinformatics*  DOI: 10.3389/fbinf.2024.1397036 (2024).
- [15] Giulia Bernardini, Estéban Gabory, Solon P. Pissis, Leen Stougie, Michelle Sweering, and Wiktor Zuba. “Elastic-Degenerate String Matching with 1 Error or Mismatch”. *Theory of Computing Systems*  DOI: 10.1007/s00224-024-10194-8 (2024).

Languages

- French: Native
- English: Proficient
- Italian: Very good